

COMMODITY VIEWS

From Energy to Metals: Why to Still Diversify Into Commodities

- As energy flows start to normalize through the Strait of Hormuz, we take stock of how commodities have performed year to date, highlighting that the energy supply shock triggered by the Iran conflict is only the most recent of several examples seen over the past year of why strategic investment portfolios can benefit from diversifying into commodities.
- Investing in different commodities can help diversify risks inherent to equity/bond portfolios under different circumstances, including (1) commodity supply shocks, like the Hormuz disruption, which might lead to higher inflation and lower economic growth, (2) structural demand support for commodities that face challenges to growing supply, and (3) a flight to real assets when fiscal sustainability or other financial risks arise. We see many of the drivers behind such circumstances, which are supportive of commodity returns, remaining relevant going forward.
- In particular, we think that the Iran conflict ultimately reinforces many of the themes supporting power and metals demand, more so than oil and gas. From a potential increased reliance on EVs, to further investment into renewable power generation, both of which require further grid investment, to potentially larger defense spending, and growing competition to win the AI race, these themes are highly supportive of power, copper, lithium and aluminum demand.
- Combined with the fact that power infrastructure can face bottlenecks and that metals refining remains highly geographically concentrated, this strong structural support to power and metals demand leaves these markets vulnerable to tightening shocks. As a result, while oil and gas have traditionally had a bigger impact on inflation, and its supply concentration and associated geopolitical risks remain, sharp price increases might start to also appear more often across power and industrial and precious metals markets going forward.

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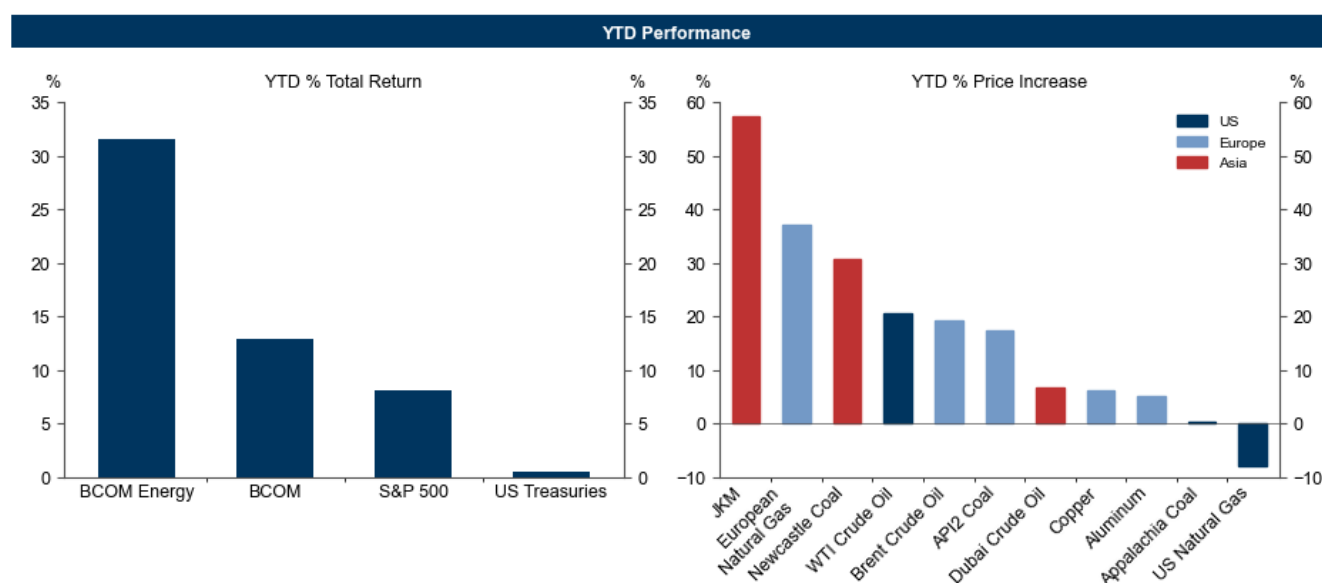
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Commodities Can Help Diversify Equities/Bonds' Risks Under a Broad Range of Circumstances

1. A large energy shock, the second in only five years. While the 2022 energy crisis impacted mostly natural gas supply, lifting liquefied natural gas (LNG) prices across Europe and Asia, this year's 16-week Strait of Hormuz disruption significantly reduced supply of both oil and gas, causing a broad increase in energy prices, while threatening the global economic outlook. Even as crude oil prices now recede following the US-Iran deal to re-open the Strait of Hormuz, broad commodity price gains year to date, combined with a strong oil roll yield¹ during the conflict, have supported year-to-date commodity returns well above those of equity and bonds (Exhibit 1), though with higher volatility.

Exhibit 1: The Strait of Hormuz disruption contributed to commodity returns outperforming other assets year to date, especially as a strong roll yield for oil combined with higher prices



Source: Bloomberg, Goldman Sachs Global Investment Research

Importantly, the upside to energy prices during the conflict was limited by what proved to be a more flexible global market than what we originally expected. In particular, the sharp drop in China liquefied natural gas (LNG) imports in Mar/Apr, the first two months of the conflict, and in oil imports to this day, helped limit global market tightness. Still, by May, three months into the conflict, crude oil and oil product (like gasoline, diesel and jet fuel) prices had rallied 43% and 63%, respectively, vs pre-war levels, while European gas and Asia LNG prices rallied 50% and 70%. And while crude oil prices have come off sharply over the past couple of weeks, the same cannot be said of gasoline or diesel, where we expect demand to rebound faster than supply, keeping balances tight for longer, and prices slower to come down.

Overall, assuming that the re-opening process continues, the Hormuz shock will likely have a limited scarring effect on global economic activity and hence on energy demand. Our economists expect global GDP growth at 2.4% this year, 0.4pp down from 2025.

¹ The roll yield is the gain or loss generated by the investor "rolling" their position from an expiring futures contract into a new contract with a later expiration date. The roll yield is positive when the expiring contract is more expensive than the futures contract they are rolling into (i.e., when the forward curve is in backwardation).

Had the Iran conflict lasted longer, its negative impact on global growth might have been significantly larger at -2pp from our pre-war expectations, underscoring the potential value of diversifying into commodities in strategic investment portfolios.

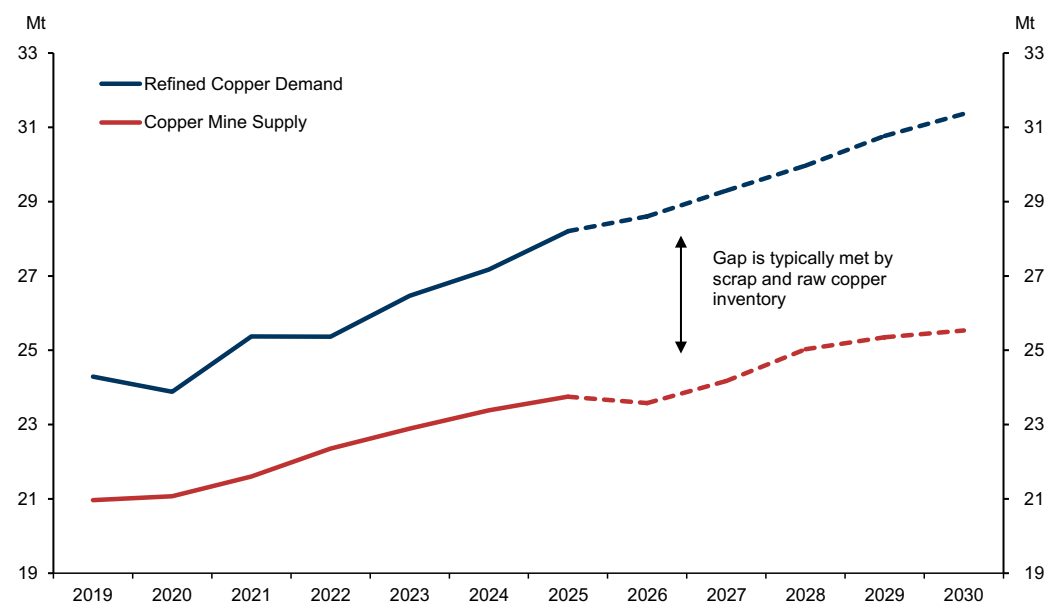
2. Long-term structural support for copper prices, with a near-term policy boost.

The pairing of strong structural support to copper demand with a long-cycle supply side, where it can take an extended period of time for supply to respond to price incentives, sets the stage for a well supported price environment for copper.

On the demand side, growth has been increasingly tied to strategic sectors, with grid and power infrastructure likely driving over 60% of copper demand growth by 2030 (vs. 2025), alongside direct boosts from defense, electric vehicles, renewable generation capacity, and data centers. Importantly, we see investment in aging Western power grids as a national security priority, due to its critical role in AI and energy security. This shifts copper demand from being mainly cyclical to increasingly strategic, making it less sensitive to economic slowdowns and high prices than traditional demand sources such as construction or white goods.

At the same time, copper supply might struggle to grow fast enough ([Exhibit 2](#)). Multiple recent [mine incidents](#) highlight the growing structural challenges in copper mining as copper mines get deeper, grades get lower and ore gets harder, requiring greater investment. This adds to operating costs and sustaining capex, and limits the ability of supply to respond quickly. As a result, copper prices have to rise to solve this gap by incentivizing higher scrap supply and potential demand substitution into alternative materials such as aluminum.

Exhibit 2: We expect copper demand growth to continue to outpace that of mined supply

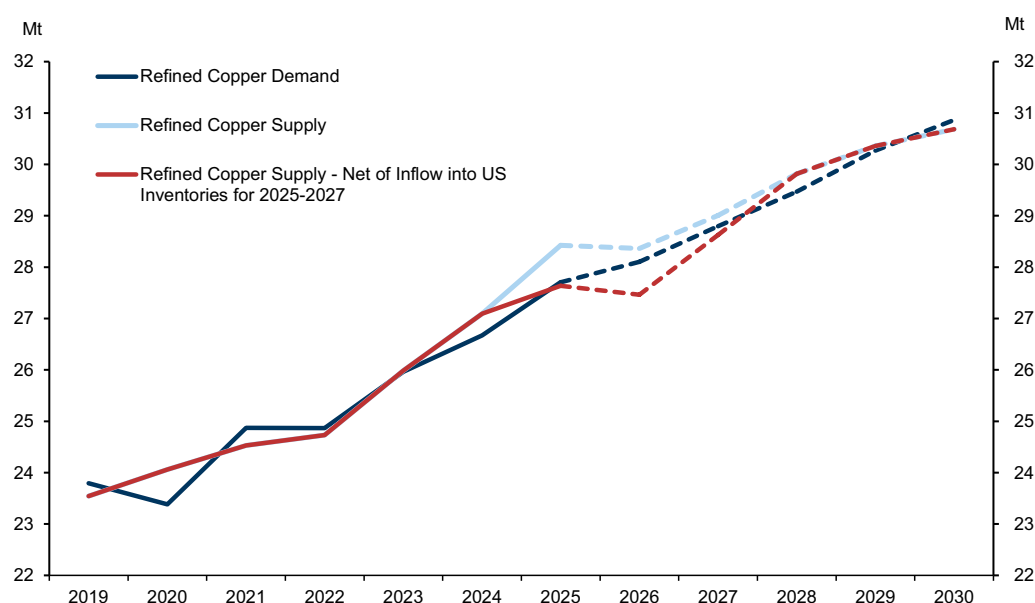


Source: Wood Mackenzie, CRU, S&P Global Market Intelligence, SMM, ICSG, Goldman Sachs Global Investment Research

While the copper balance tightening we expect is gradual in nature, shifts in policy have brought some of that tightness forward. Specifically, since the US imposed import tariffs on steel and aluminum last year, copper imports into the US picked up significantly in anticipation of a potential US copper import tariff being implemented. While there remains no US tariffs on refined² copper imports, copper tonnes have continued to head to the US, causing the ex-US copper balance to tighten significantly ([Exhibit 3](#)), and prices to briefly reach an all-time high of over \$14,000/t in May, before moderating to the low-to-mid \$13,000s.

These tighter copper balances led us to recently raise our end-2026/average 2027 LME copper forecasts to \$13,735/\$13,800. Our longer-term view remains that prices will need to rise further by the next decade to balance the market. By 2035, we think a price of \$15,000/t will be needed to keep aging mines in operation, lift scrap collection rates, deliver additional substitution and support the development of new mines.

Exhibit 3: Policy-risk-driven copper flows into the US drove the ex-US market into a deficit this year



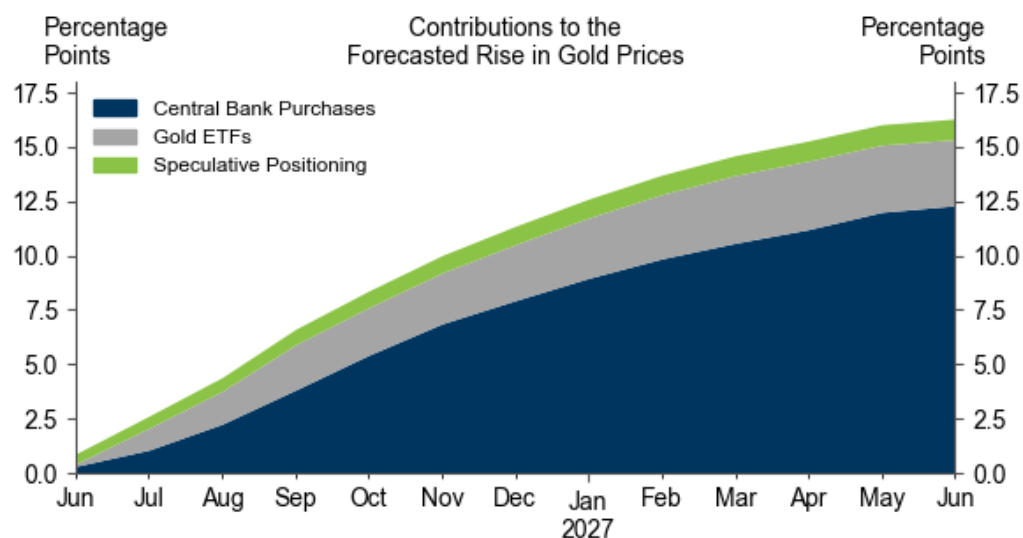
Source: Wood Mackenzie, CRU, SMM, ICSG, S&P Global Market Intelligence, Goldman Sachs Global Investment Research

3. Gold is not done. Despite gold's strong rally since 2022 (up 123%), we continue to see further upside, driven by both structural and eventually cyclical factors. Structurally, EM central bank diversification—following the 2022 freezing of Russia's reserves—remains the anchor of our \$4,900/toz end 2026 forecast ([Exhibit 4](#)). While the pace of central bank gold purchases has moderated to ~50 tonnes/month on a 3-month (seasonally adjusted) and 12-month moving average basis, we view the ongoing diversification trend as structural. A recent World Gold Council survey supports our view: a record 45% of the 76 central banks surveyed between February and May expect to increase their own gold reserves over the next 12 months, while ~90% expect global reserves to rise with the remainder expecting broadly stable holdings ([Exhibit 5](#)). We therefore assume continued central bank accumulation of 50t/month in 2026 and 40t/month in 2027.

² The US implemented a 50% Section 232 tariff on semi-finished copper products and copper-intensive derivative products effective August 1, 2025. Refined copper, ores, concentrates and scrap were excluded.

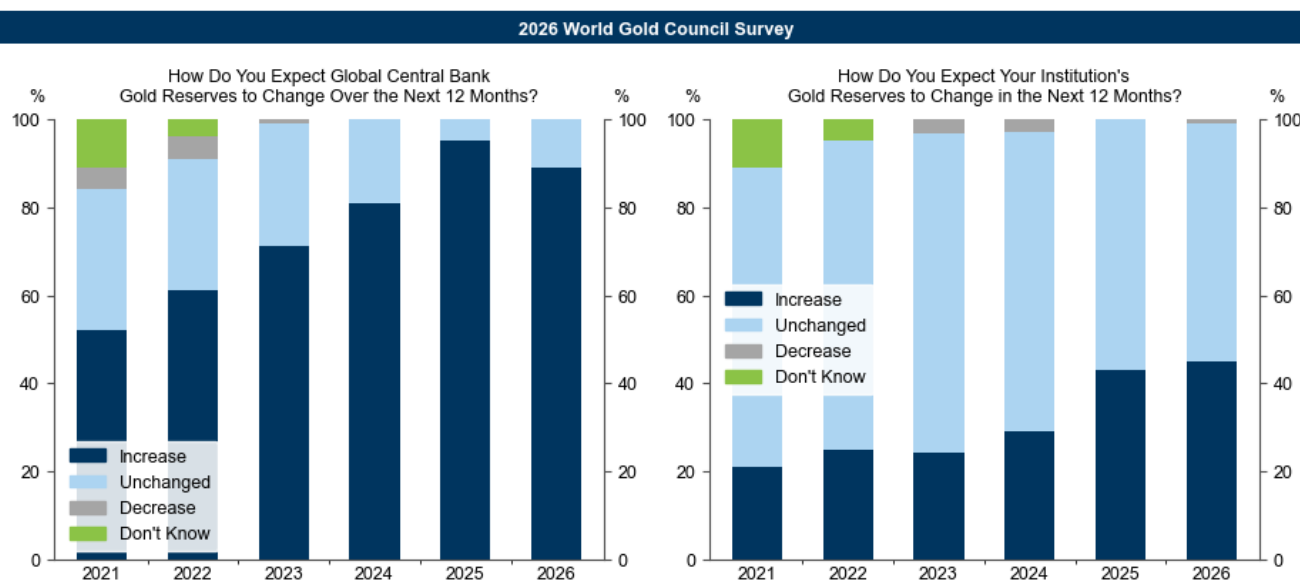
Cyclically, however, gold faces near term headwinds from the potential hit to demand for macro policy hedges (as a hawkish Fed helps fade the debasement theme) and as markets are pricing in Fed hikes this year amid inflation concerns, weighing on rate-sensitive ETF demand. We expect these headwinds to at least partly reverse over time. While we assume stable macro-policy hedges demand for gold, we expect ETF positioning to gradually rise, consistent with our economists' view of a delayed but ongoing easing cycle in 2027H2. Over the medium term, risks to our gold price forecast remain skewed to the upside on net. Gold's share in private portfolios remains small, and the Iran episode — together with broader geopolitical developments (e.g., Greenland, Venezuela) — may eventually accelerate private diversification into gold, including by weighing on perceptions of Western fiscal sustainability.

Exhibit 4: Continued central bank diversification remains the main driver of our constructive gold price outlook



Source: Goldman Sachs Global Investment Research

Exhibit 5: The recent World Gold Council central bank survey suggests continued gold demand



Source: World Gold Council, Goldman Sachs Global Investment Research

To Each (Inflation Shock) Their Own (Inflation Hedge)

While the Hormuz shock has illustrated how commodities exposure can work as a hedge against inflation, it also demonstrated with the sharp sell off in gold in the period that not all commodities work equally well as a hedge under any inflation shock. In practice, inflation typically arises through three distinct regimes – late cycle inflation, supply disruptions, and institutional credibility risk – and each calls for a different hedge.

Regime #1: Late Cycle – Hedge With Cyclical Commodities

When the business cycle runs hot, equities initially benefit from strong growth. But as the economy begins to outpace its productive capacity – what economists call a positive output gap – inflation pressures build and real bond returns weaken. Over time, rising input costs compress margins and equity growth starts to soften. It is at this point – when bond prices weaken and equity returns begin to lose momentum – that commodities tend to provide diversification through stronger returns. Commodity performance typically strengthens late in the cycle because a positive output gap implies demand exceeds supply. In commodity markets, this imbalance shows up as sustained inventory drawdowns. Late in the cycle, inventories have been drawing for long enough to approach depletion, pushing prices higher – especially for cyclical commodities like oil and industrial metals.

Regime #2: Supply Disruption – Hedge With a Broad Commodity Basket (Ex. Precious Metals)

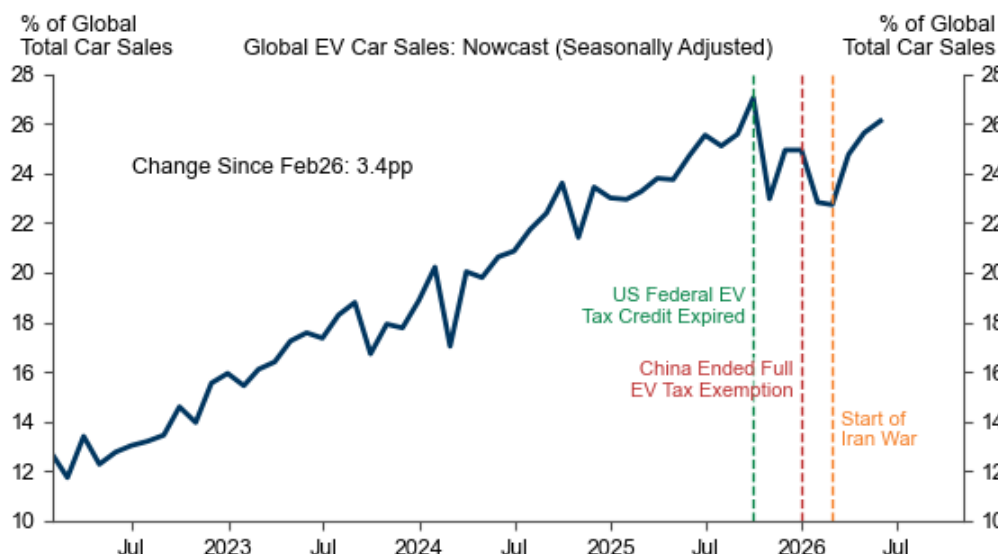
When a supply disruption hits – as just exemplified during the Iran conflict – inflation rises while growth weakens, weighing on bond and equity prices simultaneously. Commodities, as the disrupted input, are then among the few assets to deliver positive real returns. Because the source and timing of disruptions are inherently unpredictable, a broad commodity basket (ex. precious metals) offers the most robust protection.

Regime #3: Institutional Credibility Risk – Hedge With Gold

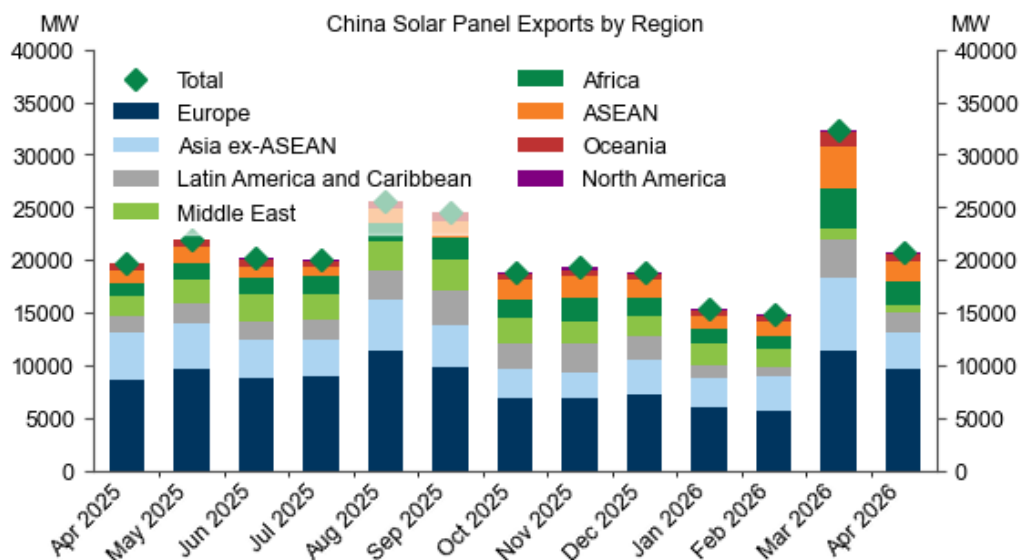
In the first two inflation regimes – late cycle inflation and supply disruptions – gold is not an effective hedge. If anything, gold often sells off initially: higher inflation can lead markets to price in rate hikes, raising the opportunity cost of holding a non yielding asset, while equity drawdowns can trigger margin call liquidations in gold, whose liquidity makes it a ready source of cash. Gold hedges a narrow inflation regime: when inflation expectations rise due to concerns around institutional credibility or macro policy, causing bonds and equities to sell off together in real terms. Gold then stands apart as the key neutral asset whose value does not rely on any government backing.

The Iran Conflict Appears on Its Way to a Resolution. What Next?

While oil demand will rebound in the aftermath of the Iran conflict, including what we expect will be an over 1 mb/d push for the rebuilding of strategic petroleum reserves, we think that the Iran conflict ultimately reinforces many of the themes supporting power and metals demand in particular. From a potential increased reliance on EVs, to further investment into renewable power generation, both of which require further grid investment, to potentially larger defense spending, and growing competition to win the AI race, these themes are highly supportive of power, copper, lithium and aluminum demand, while potentially weighing on long-term demand growth of oil and natural gas (Exhibit 6 and Exhibit 7).

Exhibit 6: Global EV sales have sharply accelerated during the Iran conflict...

Source: Government Data, IEA, Goldman Sachs Global Investment Research

Exhibit 7: ...as have China solar panel exports

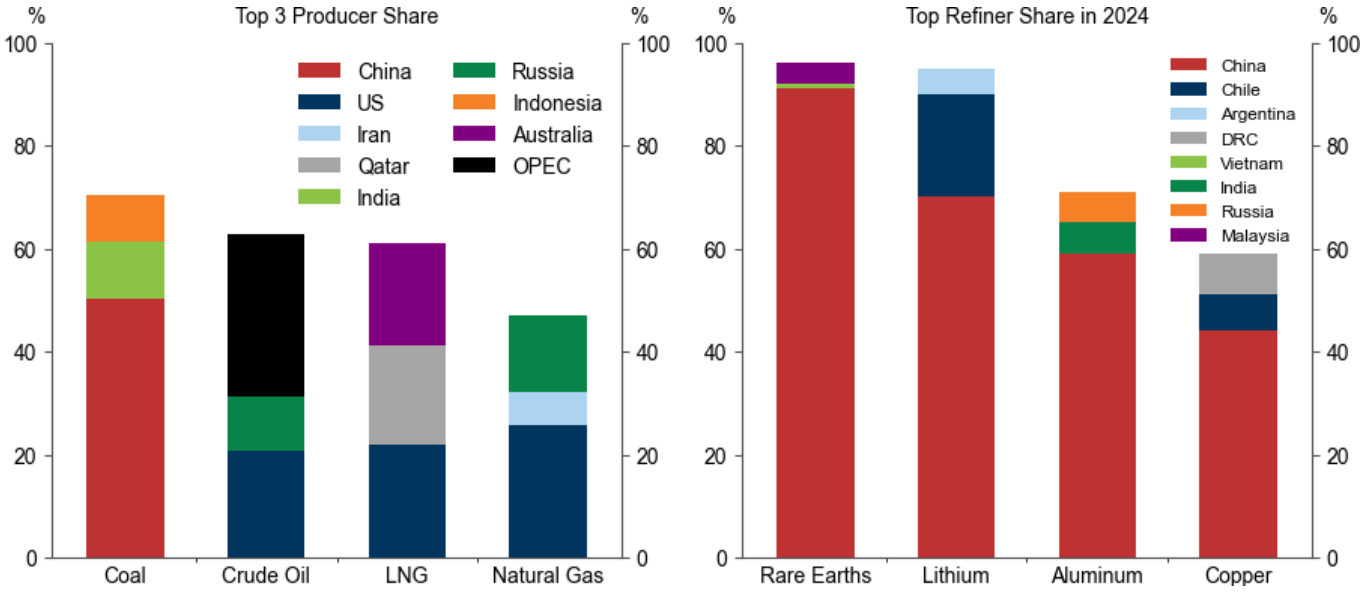
Source: Ember, Goldman Sachs Global Investment Research

Combined with the fact that power infrastructure can face bottlenecks, as has been the case in the US Mid-Atlantic power market, and that metals refining remains highly geographically concentrated (Exhibit 8), this strong structural support to power and metals demand leaves these markets vulnerable to tightening shocks.

As a result, while oil and gas have traditionally had a bigger impact on inflation, and their supply concentration and associated geopolitical risks remain, sharp price increases might start to also appear more often across power and industrial and precious metals markets going forward. In particular, our economists have flagged that higher industrial metals prices and China's export controls on rare earths and related products were among the main drivers of the recent acceleration in the US' core producer price index,

while expecting that higher US electricity prices will continue to boost consumer electricity inflation.

Exhibit 8: Commodity supply is currently heavily concentrated, with a large share of supply in geopolitical or trade-dispute hot spots



Source: EIA, IEA, WoodMac, CRU, Goldman Sachs Global Investment Research

Appendix

Exhibit 9: The recent Hormuz energy supply disruption is the latest example of why it pays to diversify into commodities

Total returns

	Dollar Weight	Historical Performance				GS Forecast	
		2023	2024	2025	YTD	Balance of Year	2027
BCOM	100	-8	5	16	15	3	1
BCOM ex. Agriculture & Livestock	63	-10	8	22	19	7	1
S&P GSCI	100	-4	9	7	26	7	1
S&P GSCI ex. Agriculture & Livestock	76	-3	11	11	37	10	0
Energy	52	-5	10	-5	55	10	-2
Industrial Metals	14	-4	3	29	7	4	-2
Precious Metals	10	12	26	69	-8	17	12
Agriculture	14	-8	0	-8	0	-3	2
Livestock	9	0	20	27	5	3	1

We highlight the commodity indices ex Agriculture & Livestock because we currently don't forecast agriculture and livestock prices, and instead use forwards.

Source: Goldman Sachs Global Investment Research

Disclosure Appendix

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We, Samantha Dart, Daan Struyven, Lina Thomas, Yulia Zhestkova Grigsby, Hongcen Wei, Lavinia Forcellese, Aurelia Waltham, Filippo Cuscito, Laura Cyr and Alexandra Paulus, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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EUROPEAN ECONOMICS ANALYST

UK—The Economic Cost of Brexit, Ten Years On

- The UK economy has materially underperformed since the 2016 EU referendum, shifting from a growth profile closer to the US to one closer to the Euro area. Our updated “Doppelgänger” analysis suggests real GDP is around 6% below its counterfactual path, more than our previous 5% estimate two years ago. While these figures could capture the impact of other factors as well as Brexit, robustness checks point to an underperformance of at least 4%. These estimates are broadly consistent with other studies. Model-based estimates of the cost of leaving the EU average around 5%, while recent empirical research finds a 6–8% impact.
- Analysis of the channels through which Brexit could have affected output also points to a significant drag. Goods trade volumes have underperformed by 10–15% since the referendum. Services trade has held up better, but recent studies nonetheless find a meaningful Brexit-related hit. Lower trade intensity has likely reduced productivity, lowering GDP by 2–4%. Business investment has also been weak since 2016—with business surveys suggesting this is partly due to Brexit—likely lowering output by 2%. The impact on labour supply through migration has been more mixed, with lower EU immigration but higher non-EU inflows. That said, shifting migration patterns have probably contributed to labour shortages in some sectors.
- Taken together, our estimates indicate that Brexit has materially weighed on Britain’s economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.
- That said, we see two reasons why the peak Brexit headwind might now be behind us. First, estimates point to a front-loaded drag, suggesting that the Brexit growth headwind should ease over time. Second, opinion polls now show greater support for a closer EU relationship, and the government is pursuing more EU-UK cooperation. An ambitious reset—such as closer single-market or customs-union alignment—could reverse a meaningful share of the Brexit cost and support stronger trend growth. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

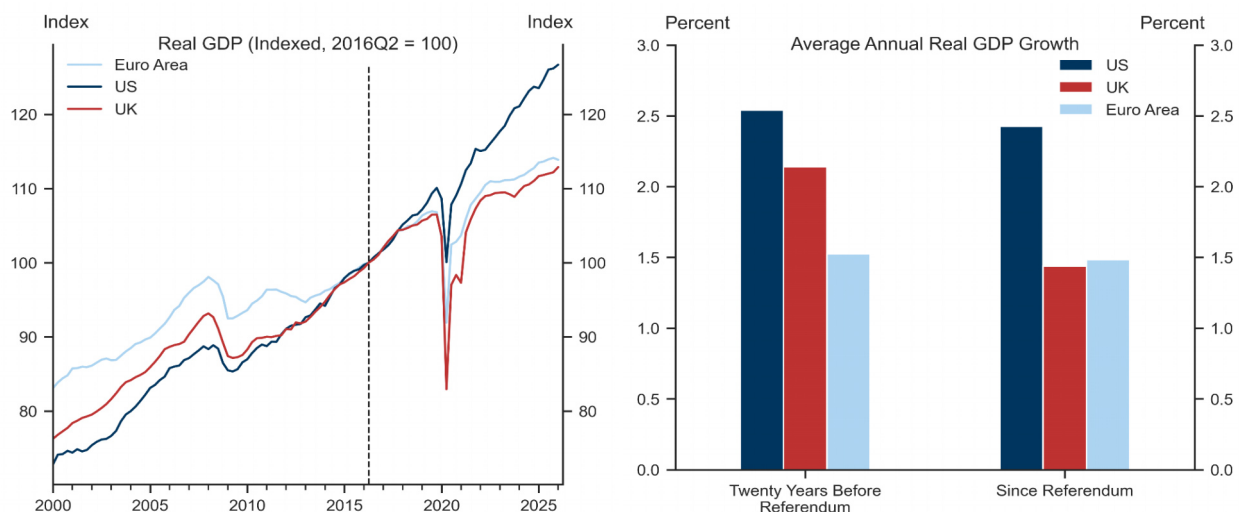
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UK—The Economic Cost of Brexit, Ten Years On

The UK voted to leave the EU ten years ago this week and has underperformed other advanced economies significantly since then (Exhibit 1). UK growth looked more similar to the US in the twenty years leading up to the referendum, but more similar to the Euro area in the subsequent ten years.

Exhibit 1: UK Growth Was Closer to the US Before Brexit, More Like the Euro Area Since Then



Source: Goldman Sachs Global Investment Research, Haver Analytics

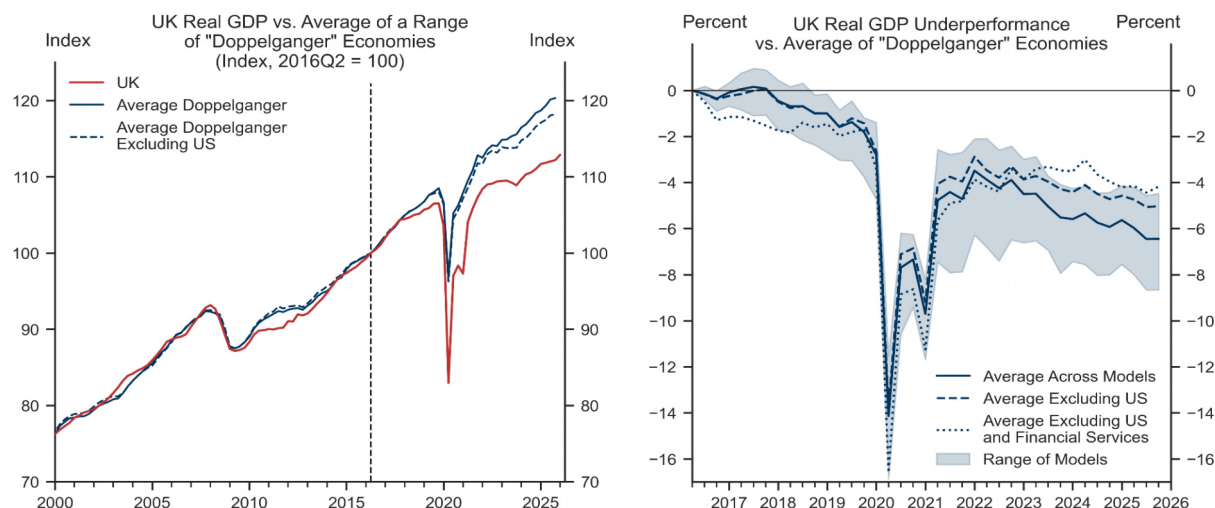
Given the ten-year anniversary of the referendum, we take stock of the macroeconomic costs of the Brexit vote and update our 2024 estimates.

Counting the Cost

We start by examining empirically how much UK real GDP has underperformed other advanced economies since 2016 using a “Doppelgänger” analysis.¹ This method uses statistical techniques to find the best combination of other countries that match the path of UK real GDP before the referendum and uses this same combination to project what might have happened thereafter.

Our estimates suggest that the UK has fallen short of its Doppelgänger by 6% since the referendum, compared with the 5% we estimated two years ago (Exhibit 2, left). Varying the group of countries in the analysis—including the G7, G10, OECD, OECD Europe—points to a range of 4–8% for the output cost (Exhibit 2, right).

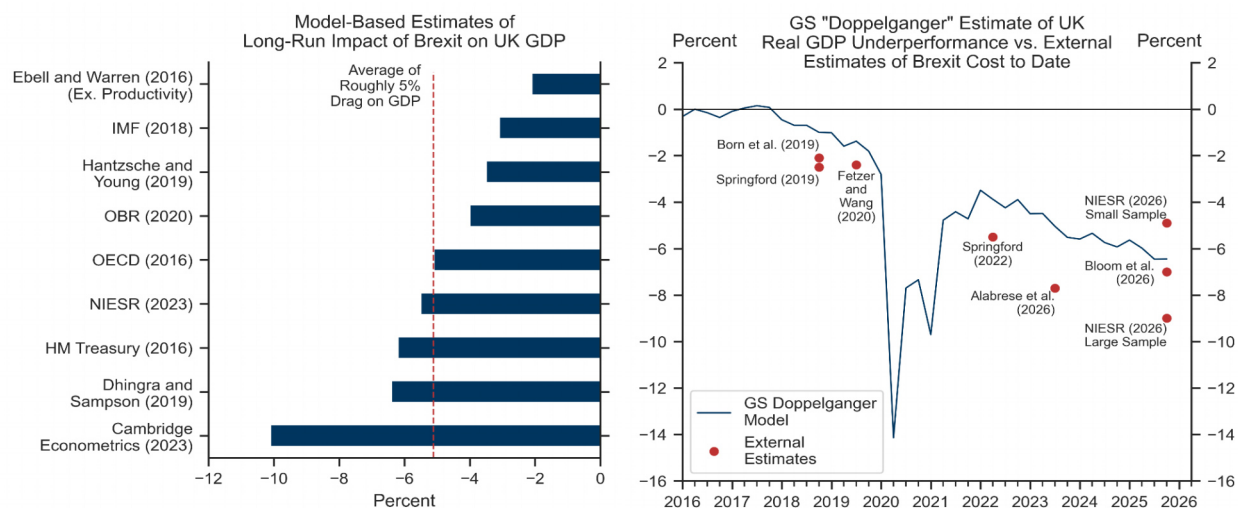
¹ This synthetic control method was applied to Brexit by Benjamin Born and co-authors in “The Costs of Economic Nationalism: Evidence from the Brexit Experiment,” May 2019.

Exhibit 2: Our Doppelgänger Model Points to Significant Underperformance Since the Referendum

Source: Goldman Sachs Global Investment Research, Haver Analytics

A shortcoming of the Doppelgänger approach is that not all of the post-2016 underperformance can be attributed to Brexit, as other structural changes have affected Britain's relative economic performance. In particular, the inclusion of the US could skew the estimates, as the US has benefited from a much larger technology boom than other advanced economies during this time. Excluding the US from the Doppelgänger analysis points to a smaller output cost at 5% (dashed line in Exhibit 2, right). It is also possible that UK growth was temporarily boosted by a booming financial sector before the referendum relative to other advanced economies. Re-running the analysis on UK GDP without the financial sector and again excluding the US points to 4% underperformance, which is likely to be a lower-bound estimate (dotted line in Exhibit 2, right).

Our Doppelgänger estimates are consistent with prior studies for the economic cost of Brexit. Model-based studies—which simulate the economic repercussions of greater trade frictions outside of the EU—point to a total output cost of Brexit of around 5% (Exhibit 3, left). The latest empirical studies—often using Doppelgänger models—point to a hit of 6-8%, which has grown notably in recent years (Exhibit 3, right). We provide a more detailed summary of these studies in the appendix.

Exhibit 3: Studies Point to a Large (and Rising) Cost of Brexit

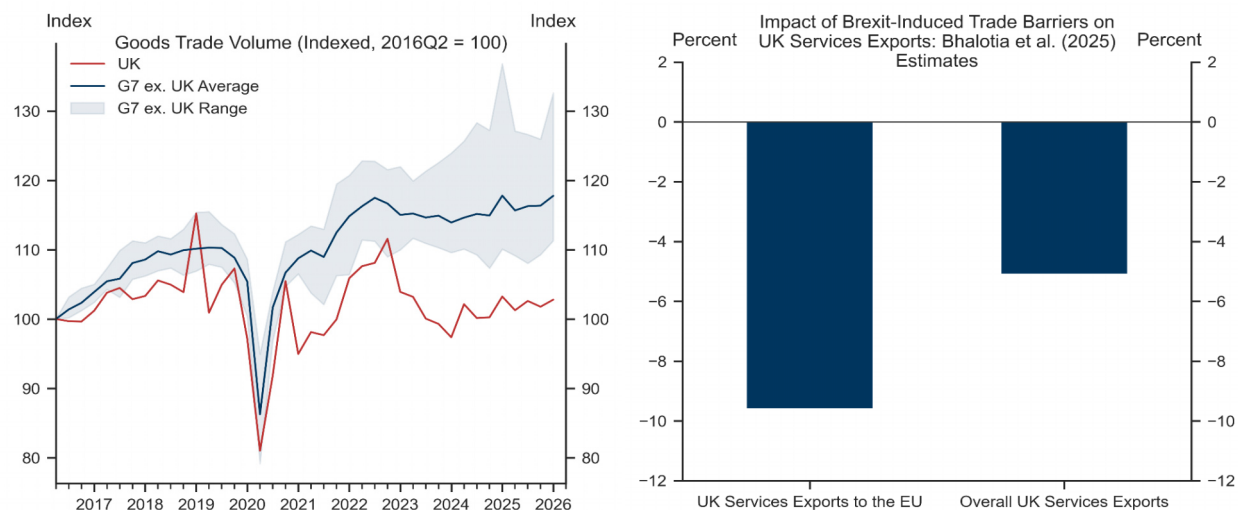
Ebell and Warren (2016) note that incorporating productivity losses would substantially increase their estimates.

Source: Goldman Sachs Global Investment Research, Various academic studies

Assessing the Channels

Given uncertainty around the top-down estimates, we consider the mechanisms of Brexit to help gauge how much of the UK's underperformance can be attributed to Brexit. Studies generally consider three mechanisms through which Brexit could have lowered long-term output: (1) international trade, (2) business investment and (3) immigration.

First, the UK has materially underperformed other advanced economies in terms of trade since 2016. We estimate that goods trade volumes would be 10-15% higher in a counterfactual in which volumes had grown in line with the average of other G7 economies, reflecting higher trade frictions outside of the EU (Exhibit 4, left). Services trade has held up better. Recent studies, however, also point to a significant hit to services trade as a result of leaving the EU (Exhibit 4, right).

Exhibit 4: UK Trade Has Been Hit

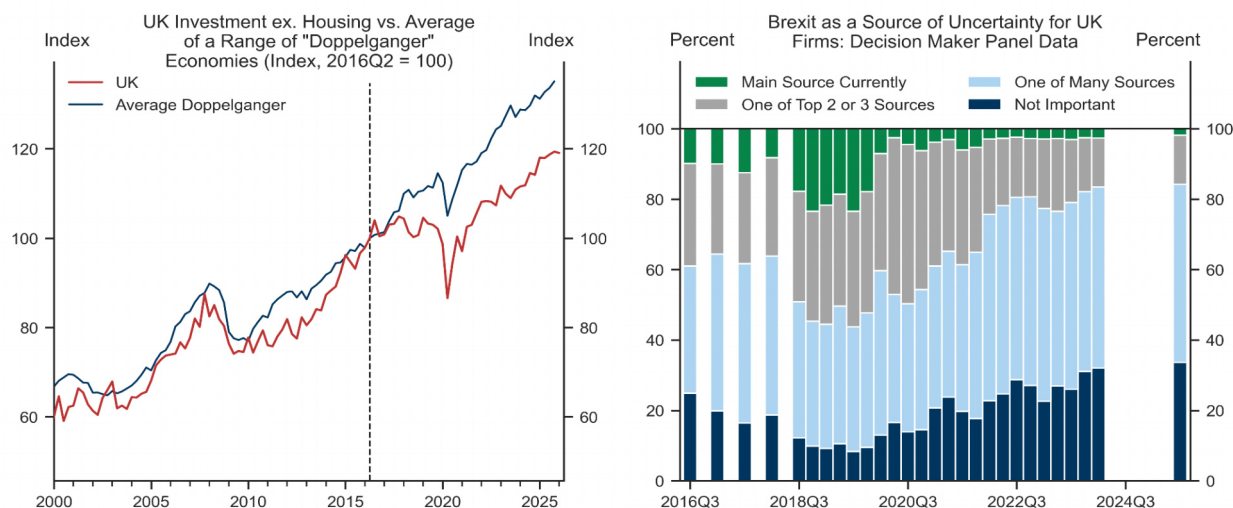
Source: Goldman Sachs Global Investment Research, Haver Analytics, Bhalotia et al. (2025)

Overall, the trade estimates point to a roughly 5-10% decline in trade volumes due to Brexit. Previous studies suggest that a decline in trade intensity of this magnitude could have reduced GDP by 2-4% through reduced productivity.²

Second, UK investment growth stalled after the referendum, falling significantly below its pre-2016 trend (Exhibit 5, left). Our Doppelgänger estimates for investment point to a shortfall of around 10% relative to other economies. At least some of this underperformance likely reflected Brexit uncertainty after the referendum, with surveys clearly flagging Brexit as a source of uncertainty (Exhibit 5, right). Our estimates on the business investment channel imply an output hit of just over 2% by end-2025.³

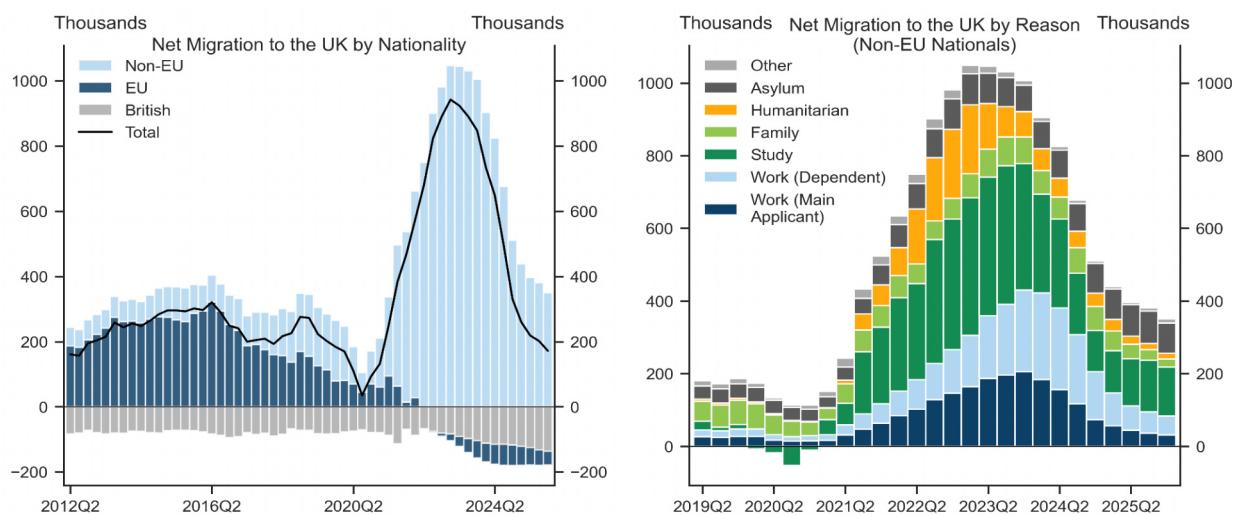
² Two studies by Feyrer using different methodologies have estimated that each 1% reduction in trade lowers GDP by around $\frac{1}{4}\%$ or $\frac{1}{2}\%$.

³ Haskel and Martin (2023) estimated that weakness in business investment after the referendum had lowered output by 1.3% by 2022.

Exhibit 5: Investment Has Underperformed

Source: Goldman Sachs Global Investment Research, Haver Analytics, Bank of England

Third, Brexit has affected immigration flows into the UK. Net migration from the EU has fallen sharply from a peak of over 300k in the year to 2016Q2 to -42k in 2025 (Exhibit 6, left). At the same time, net migration from outside the EU rose significantly to over 1m per year after the pandemic. Although much of the pickup in non-EU immigration reflects reasons other than work (Exhibit 6, right), recent [research](#) suggests that the new immigration system has so far led to a modest net increase in the number of foreign-born people in work. That said, net migration has dropped sharply recently, partly reflecting new government policies.

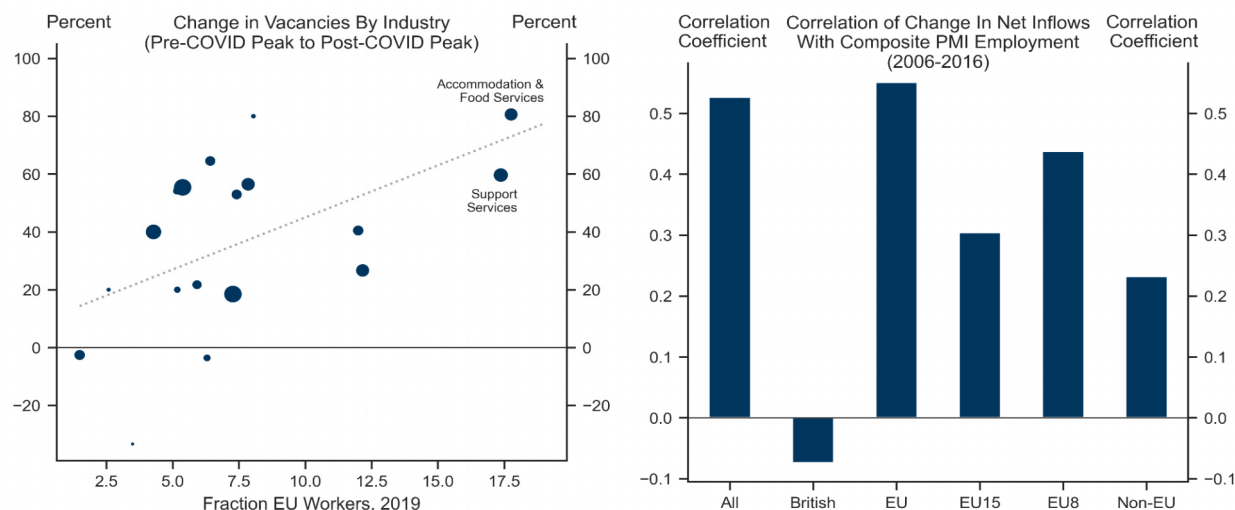
Exhibit 6: Net Migration from the EU Has Fallen but Overall Inflows Rose Post-Pandemic

Source: ONS

Despite the rise in net migration after Brexit, our previous [work](#) suggests that changes in the immigration system have probably contributed to labour shortages. EU nationals have historically been concentrated in lower-skilled occupations and are harder to replace under the new skill-selective migration regime. The sectors that are most reliant on EU workers have experienced the largest increases in vacancy rates since the Covid

pandemic (Exhibit 7, left). Going forward, reduced cyclical volatility of migration flows compared with the pre-Brexit period (Exhibit 7, right) could lead to greater volatility in labour market tightness and domestic inflationary pressures.

Exhibit 7: Lower Immigration from the EU Helps Explain Labour Shortages



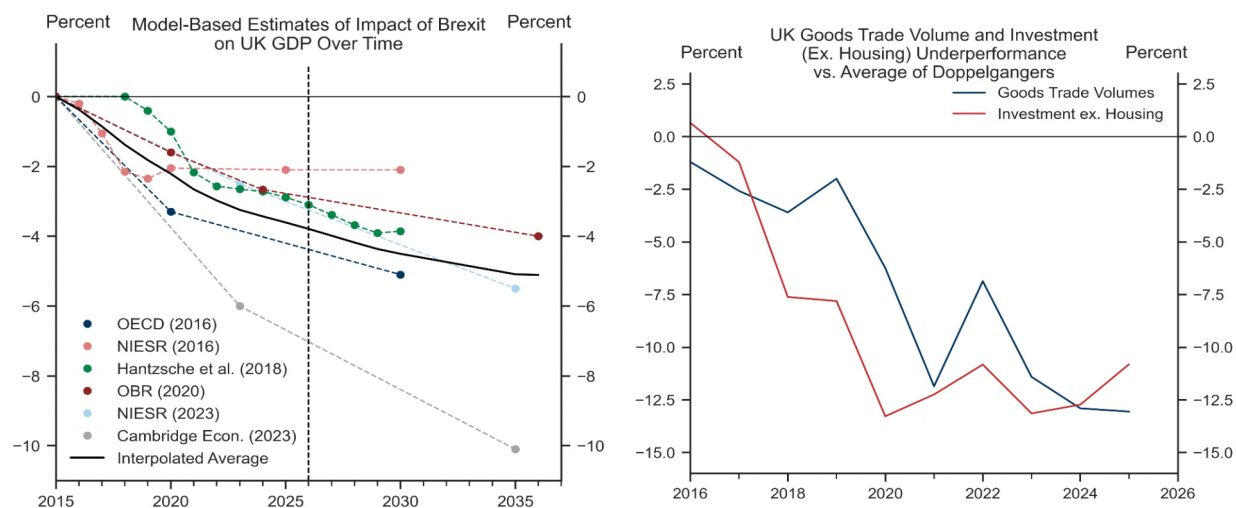
Source: Goldman Sachs Global Investment Research, Haver Analytics, ONS

As Bad as It Gets?

Taken together, our estimates show that Brexit has materially weighed on Britain's economic performance relative to other advanced economies. While new trade agreements could, in principle, help mitigate some of the costs of Brexit, the benefit from new (non-EU) trade deals is likely to be small.

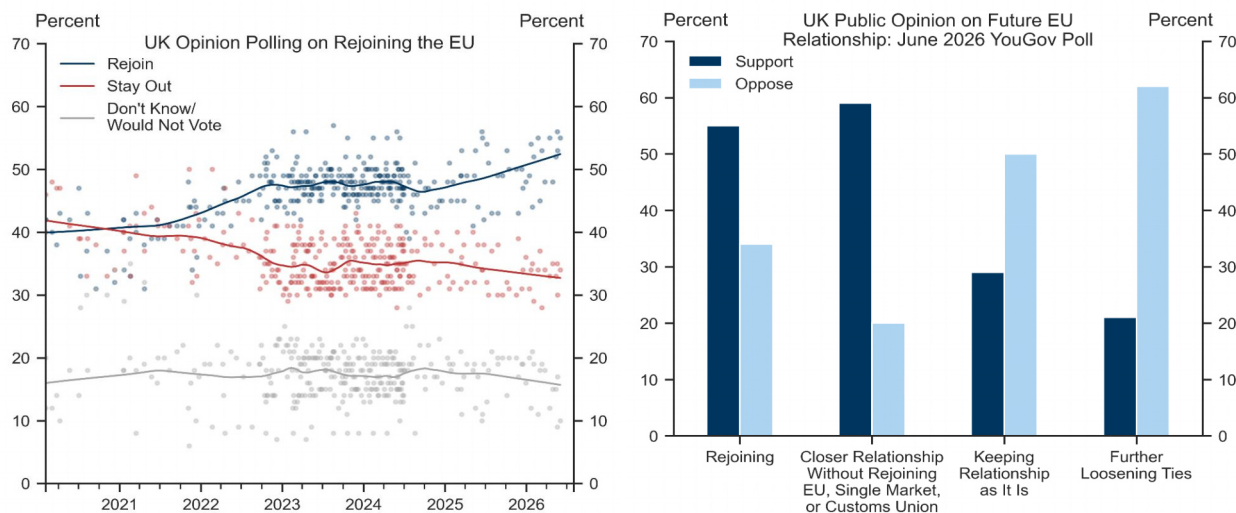
That said, we see two reasons to think that the worst of the Brexit headwind might now be behind us.

First, although the full impact of leaving the EU is likely to take time to materialise, several studies point to a front-loaded growth drag (Exhibit 8, left). Decision Maker Panel data and our own analysis indicate that much of the impact on investment had already come through before the UK left the EU at the end of 2020 as businesses anticipated future disruption to trade barriers. Although there is evidence that trade flows can take 7-10 years to adjust fully to new trade barriers, the shortfall in goods trade volumes relative to other economies shows some signs of levelling off in the last two years (Exhibit 8, right). It is true that the UK's underperformance on GDP has continued to build, perhaps reflecting lags from lower investment and trade into productivity. But these factors suggest that the growth hit is likely to gradually ease over time.

Exhibit 8: Past the Peak Brexit Growth Headwind

Source: Goldman Sachs Global Investment Research, Haver Analytics, Various academic studies

Second, recent opinion polls and political shifts point to building support for a closer EU-UK relationship. Public opinion polls show that a majority of the UK population now support rejoining the EU (52%), up notably since Covid and now comfortably ahead of the 33% share supporting Brexit (Exhibit 9, left). Public support is highest for a closer relationship with Europe without rejoining the EU, Single Market or Customs Union (Exhibit 9, right). As a result, the government has worked towards a “reset” in EU-UK relations and Andy Burnham (who is likely to succeed Prime Minister Starmer) has historically supported closer integration with Europe.

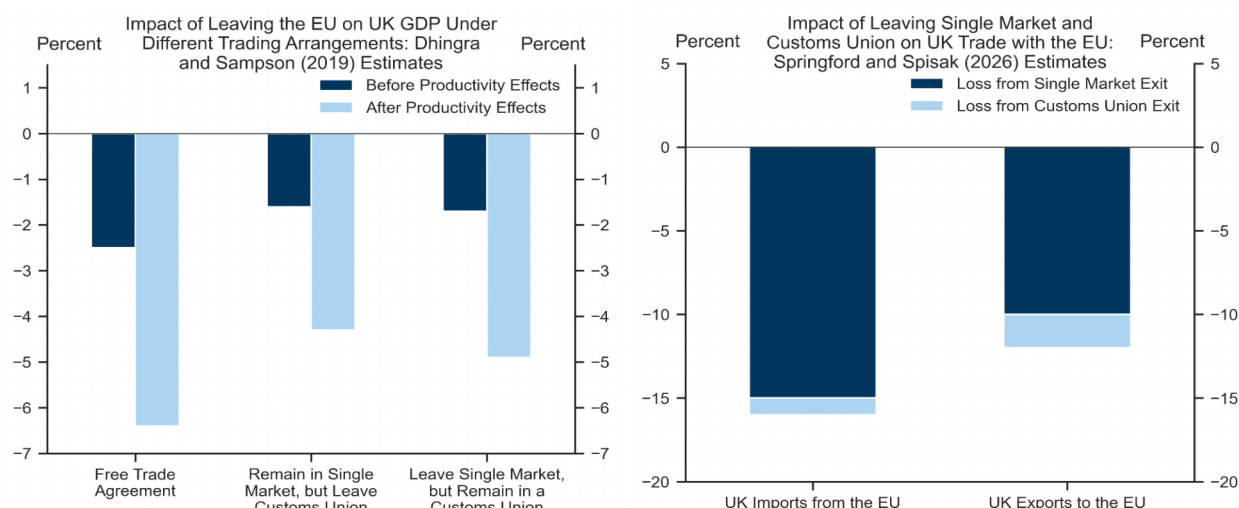
Exhibit 9: Public Opinion Has Shifted in Favour of a Closer Relationship

Source: Goldman Sachs Global Investment Research, Various polling companies, YouGov

The government could start by finalising the veterinary agreement, negotiating mutual recognition of professional qualifications, and aligning with EU rules where it is of clear economic benefit (e.g., for chemicals or the automotive sector). A more ambitious reset further down the line—such as rejoining the single market or negotiating a customs union—could raise GDP by 1-2% and thereby undo a meaningful share of the economic

cost incurred by Brexit (Exhibit 10).

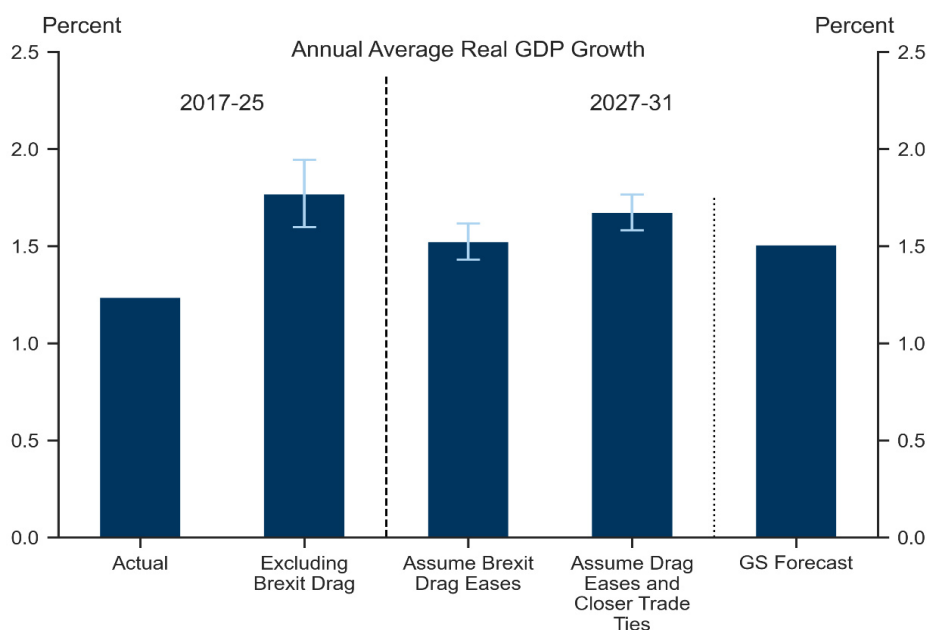
Exhibit 10: A Closer Economic Relationship Could Mitigate Some of the Brexit Costs



Source: Dhingra and Sampson (2019), Springford and Spisak (2026)

The combination of a receding Brexit headwind and a closer economic relationship with the EU could improve the UK's growth performance significantly in coming years (Exhibit 11). Simple calculations suggest that—assuming an unchanged underlying growth pace—UK trend growth could pick up to 1.5% in coming years with a diminishing Brexit drag. That figure could rise to 1.7% in the event of a significant reset of EU-UK relations.

Exhibit 11: An Easing Brexit Drag Could See Trend Growth Improve



“Excluding Brexit drag” uses the average of the GS doppelganger estimate and the growth headwind implied by the model-based estimates shown in Exhibit 8 (left). The light blue bracket shows the range from those two approaches. “Assuming Brexit Drag Eases” assumes that the drag fades in line with the time profile from the model-based estimates. “Closer Trade Ties” assumes a boost of 1-2% from a more ambitious reset over 10 years.

Source: Goldman Sachs Global Investment Research, Haver Analytics

While there is considerable uncertainty around the exact timing and magnitude of that improvement, the likely scale of the drag on growth over the past decade and the potential upside from a closer EU-UK relationship suggests that the upside is material. Consistent with this, we expect average growth in the UK to pick up from 1.2% since 2016 to 1.5% over the coming years.

James Moberly

Sven Jari Stehn

Appendix

Exhibit 12: Studies Estimating the Impact of Brexit on UK GDP

Model-Based Estimate of Long-Run Impact of Brexit			Tracking Estimates of the Impact of Brexit on GDP to Date		
HM Treasury	2016	6.2% under a bilateral agreement over 15 years	Born et al.	2019	1.7-2.5% by the end of 2018
OECD	2016	3.3% by 2020, 5.1% by 2030	Springford (CER)	2019	2.5% by the end of 2018
NIESR	2016	2.1% under FTA ex. productivity; productivity effects could increase cost significantly further	Fetzer and Wang	2020	2.3-2.5% aggregate impact by 2019, 8.5% in worst affected regions
Dhingra et al. (LSE/CEP)	2017	1.3-2.7% ex. productivity; 6.3-9.4% inc. productivity	Springford (CER)	2022	5.5% by 2022Q2
IMF	2018	2.6-3.9% under FTA, average of 3.1%	Alabrese et al.	2026	7.7% by 2023
Hantzsche et al. (NIESR)	2018	3.9% under May deal	Bloom et al.	2026	6-8% by end of 2025
Dhingra and Sampson (LSE/CEP)	2019	2.5% ex. productivity, 6.4% inc. productivity under Johnson deal	NIESR	2026	4.9-9.0% by end 2025, depending on comparison group of countries
Hantzsche and Young (NIESR)	2019	3.5% under Johnson deal			
OBR	2020	4% in the long-run under an FTA			
NIESR	2023	2-3% by 2023, 5-6% by 2035			
Cambridge Econometrics	2023	6.0% by 2023, 10.1% by 2035			

Source: Data compiled by Goldman Sachs Global Investment Research, Various academic studies

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